

Smart Farming & Precision Agriculture in Kenya

Innovation in the IT Field: Technology transforming agriculture

Capstone Project

BBIT

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What Is Smart Farming?

- Use of digital technology in farming
- Monitors crops & soil conditions
- Helps plan decisions accurately
- Improves farming efficiency
- Uses sensors, drones, data tools



Agriculture in Kenya

- Major sector for economy
- Source of food & income
- Employs large portion of population
- Faces climate & resource challenges



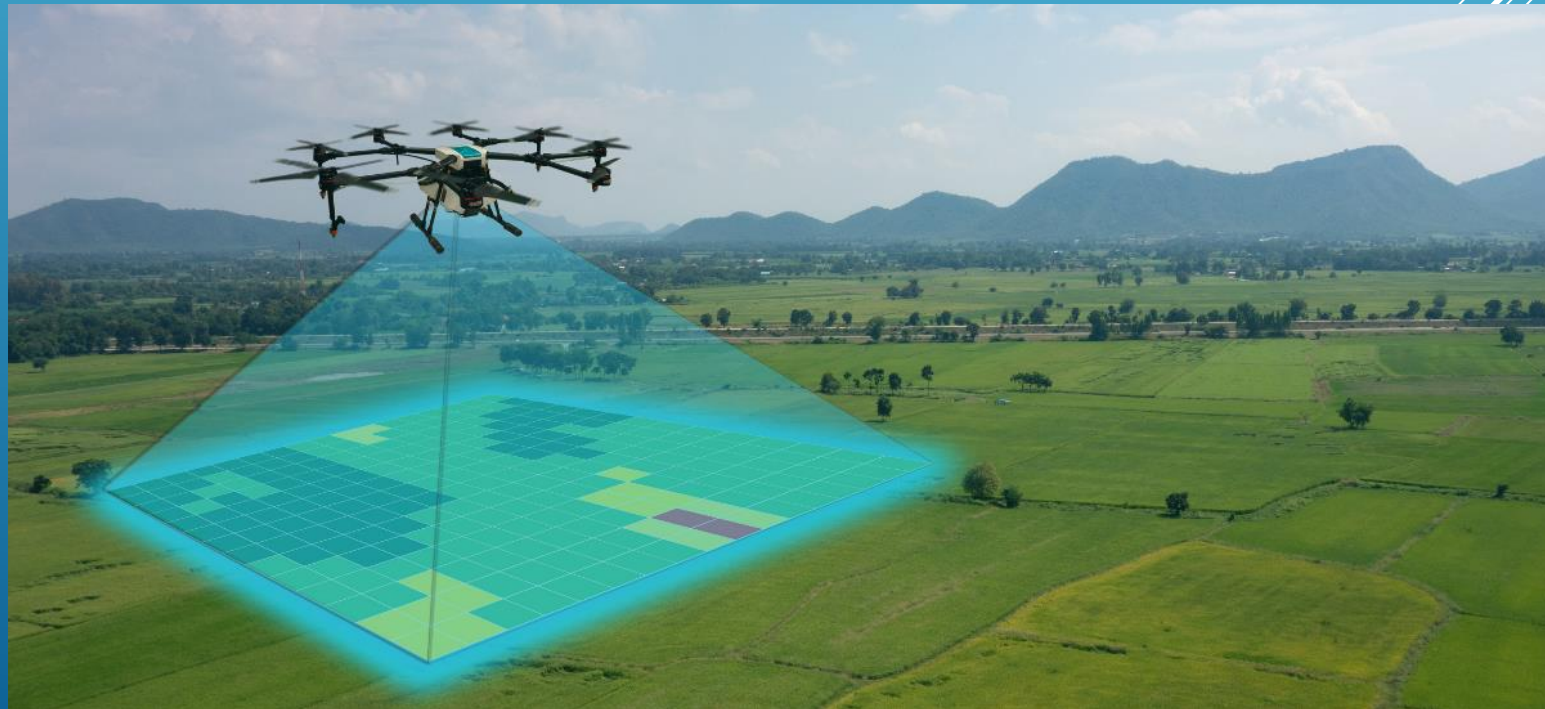
Key Technologies

- IoT soil & moisture sensors
- Drones for crop monitoring
- GPS & satellite imagery
- Mobile apps & analytics



IoT & Sensors

- Real-time soil data
- Moisture & nutrient monitoring
- Improves irrigation planning
- Saves water & inputs



Drones in Farming

- Aerial crop inspection
- Detects pest & disease early
- Helps plan spraying tasks
- Reduces time and cost



Data & Decisions

- Data collected from tools
- Analyzed by analytics software
- Provides farming insights
- Guides irrigation & input use



Impact in Kenya

- Better crop productivity
- Less waste of resources
- Faster problem detection
- Early decision support



Kenyan Examples

- Tech adoption by farmers
- Sensor systems in farms
- Drones used for surveys
- Mobile tools connect farmers



Benefits of Smart Farming

- Increased crop yields
- Efficient resource use
- Reduced environmental impact
- Better market data access



Challenges in Smart Farming

- High equipment costs
- Limited rural internet
- Need for tech training
- Adoption is still growing



Growth Opportunities

- Mobile-based tools expansion
- Government support programs
- Agritech startups emerging
- Youth involvement increases



Youth & Tech

- New jobs for youth
- Data tech & support roles
- Skills grow with farming tech
- Encourages innovation



Food Security

- Stable crop production
- Efficient use of water & inputs
- Early problem identification
- Supports community food needs



Conclusion

- Smart farming boosts productivity
- Tech improves decision-making
- Kenya benefits from innovation
- Continued growth is possible



Thank you for listening

